Top Achievements

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Purpose

This note distils the strongest evidence from my CV for promotion, fellowship, and panel discussions. Items are ranked by **impact**: funding scale, venue prestige, independence, translation to real-world use, and external recognition.

Overall ranking — top 14

1. **Horizon Europe Marie Skłodowska-Curie Postdoctoral Fellowship (CASREx).** Grant No. 101205678; EUR 276,187.92; 24 months; Imperial College London. I lead an independent fellowship on cable-driven exosuits for upper-arm muscle-impairment rehabilitation.
2. **Research independence across three domains.** Soft medical robotics (PhD, Auckland) → cable-driven robots (CUHK) → garment robotics with imitation learning (TransGP) → rehabilitation exosuits (Imperial CASREx).
3. **Industry-funded translation — BvW Holding stent testing on RoSE.** NZD 39,000 industry contract (2018); outputs in *Soft Robotics* (2021) and *IEEE TIE* (2021).
4. **Flagship journal pipeline (first-author core).** *Mechanism and Machine Theory* (2025); *IEEE TCST* (2023); *Soft Robotics* RoSE (2021); *IEEE TIE* (2020, 2021).
5. **Recent high-profile acceptances in manufacturing robotics.** *IEEE T-ASE* (2026, fabric alignment); *IEEE IROS* (2026, RTFF, equal contribution); *IEEE RA-L* (2025, equal contribution).
6. **Led multidisciplinary garment-robotics programme at TransGP** (Jul 2024–Oct 2025). Dual-arm fabric testbeds; mesh perception + imitation learning; contributions to sewing, flattening, and destacking workflows.
7. **European Talent Academy 2026** (Imperial College London, TU Munich, Politecnico di Milano). Selected Dec 2025; Munich session completed Jun 2026.
8. **Invited speaker, World Robotics Conference 2025, Beijing.** Talk: *Fabric modeling and perception for garment manufacturing*.
9. **PhD thesis output density** (Auckland, 2016–2021). RoSE platform; 3 journal articles, 2 conference papers, 1 Springer book chapter (2019).
10. **Postdoctoral research track record at CUHK** (2021–2024). MMT and TCST papers; grant co-authorship; mentoring of UG and PhD researchers.
11. **Peer review service** (2021–present). Reviewer for *IEEE RA-L*, *ICRA*, and *ASME JMR*.
12. **Best Conference Paper Award, RiTA 2017**, Daejeon, Korea.

13. **GATE 99.12% percentile** (Rank 1,211 / 137,853), 2011.

14. **Three years lecturing** (Assistant Professor Grade-1), Galgotias University, 2013–2016.

By framework category

Funding & grants

- MSCA Postdoctoral Fellowship — **EUR 276,187.92** (lead)
- BvW Holding RoSE stent testing — **NZD 39,000** (lead)
- ITF PRP and RTH-ITF grants — co-author (CUHK)
- PhD scholarships — University of Auckland; Riddet CoRE; MHRD GATE (IIT)

Research outputs (quality & quantity)

Journals: IEEE T-ASE (2026, accepted); IEEE RA-L (2025); MMT (2025); IEEE TCST (2023); Soft Robotics (2021, RoSE & SoGut); IEEE TIE (2020, 2021).
Conferences: IROS 2026 (accepted); IEEE ICMA 2025; ICRA 2023 poster; M2VIP 2017/2018; Springer book chapter (2019).

Research independence & leadership

- PI/lead — MSCA CASREx fellowship project
- Programme lead — TransGP fabric manipulation (2024–2025)
- Assistant Supervisor — 2 MEng placement students, Imperial (2026)
- Mentoring — UG FYP and PhD students at CUHK; GTA at Auckland
- Workshop organiser & presenter — IEEE ICMA 2025

Impact — academic

Invited talks (Imperial Dept. Seminar, Robotics PG Committee, WRC 2025); IEEE reviewer; RiTA best paper (2017); ETA 2026 selection; GATE 99.12%.

Impact — social & economic

Stent testing for implant manufacturers (BvW); garment manufacturing automation (sewing, alignment, flattening); upper-arm rehabilitation exosuits (CASREx); medical simulators (RoSE, SoGut).

Collaborations & network

Imperial (Morph Lab); TransGP/HKU (Prof. Kosuge); CUHK (Prof. Lau); Auckland (Prof. Xu, Prof. Cheng); LS2N Nantes (Prof. Caro); multi-country track: India → NZ → Hong Kong → China → UK.

Teaching & supervision

Guest lecture — PyDrake simulation & control, DESE71002, Imperial (2026); MAEG5090 Topics in Robotics, CUHK (50 students); 4 Auckland UG courses as GTA; 9 UG + 1 PG courses as Lecturer, Galgotias.

Elevator top 5 — for a 2-minute panel

1. **I lead a Horizon Europe Marie Skłodowska-Curie Fellowship (CASREx, EUR 276k) at Imperial,** developing cable-driven exosuits for upper-arm muscle-impairment rehabilitation with human trials, EMG, and digital twins.
2. **I have a sustained publication record in top robotics and control venues:** *Soft Robotics*, *IEEE TIE*, *IEEE TCST*, *MMT*, *IEEE RA-L*, *IEEE T-ASE*, and *IROS* — spanning healthcare, cable robots, and garment manufacturing.
3. **My work translates beyond academia:** industry-funded esophageal stent testing (BvW, NZD 39k → RoSE platform), and leading fabric-robotics R&D at TransGP for real garment-manufacturing workflows.
4. **I lead research, not only contribute:** MSCA project PI; led the TransGP fabric programme; now Assistant Supervisor for two MEng students; mentor and grant co-author from CUHK onwards.
5. **I am gaining European and international recognition:** European Talent Academy 2026; invited speaker at World Robotics Conference 2025; active reviewer for RA-L, ICRA, and JMR.

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